PARSONS

Received dar 10 OCT 2023

MBL MODERN BUILDING LEADERS ( MBL) Co. LLC شركة قادة البناء الحديث المحدودة

# Work Inspection Request

| | | | | | |
|---|---|---|---|---|---|
| **Request Ref.:** | L01-MBLC-CON-ELE-WIR-004002 | **Revision:** | R00 | **Submission Date:** | 09/10/23 |
| **Project:** | L01/ 01 | **Inspection Date:** | 10/10/23 | **Time:** | 10:00 |
| **Client:** | CPPA | **Project/Zone No:** | L01/15 | | |
| **PMCM/Engineer:** | Saudi Arabian Parsons Ltd | **Contractor:** | MBL | | |

**Discipline:**

☐ Survey ☐ Civil ☐ Architectural ☐ Structural ☐ Landscape
☒ Electrical ☐ Low current ☐ Mechanical ☐ HSE ☐ Testing/Commissioning
☐ Others (.....) Sub- discipline (.....)

**Description:**
Inspection request for wire pulling for SPS system loop (2nd fix) at MOB building First floor zone-15 in Room FF-Z5-120,FF-Z2-01 area as per attached highlighted drawings. Cu / PVC -4mm2 wire -60M FCU OUTLET-1NOS,DDC OUTLET-2NOS,30A DISCONNECT SWITCH-1NO

| | | | |
|---|---|---|---|
| **Approved Method Statement:** | L01-MBLC-CON-LVS-MES-000124-R02 | **Section:** | NN |
| **Approved Inspection and Test Plan:** | L01-MBLC-CON-ELE-ITP-000724-R01 | **Clause:** | NN |
| **Clearance Form Enclosed:** | ☐ Yes / Disciplines (.....) | | ☒ No |

| | | | |
|---|---|---|---|
| **Approved Drawing No.:** | L01-MBLC-D04-SPS-DWG-10BD102-01-LAY402-R08 | Attached Approved Drawings | ☒ |
| **Location/Sub location / Level / Room Name & No. / Grid Line:** | **Zone – 15 (First Floor)** | Attached Location Sketch | ☒ |
| | | Attached Check Sheet | ☒ |
| | | Attached Previous Revision (if any) | ☐ |

**The above work has been completed according to Project requirements and specifications and is ready for inspection**

**MBL Constriction's Representative:**

| | | | | | |
|---|---|---|---|---|---|
| **Name:** | for Yasin Sara | **Signature:** | | **Date:** | 09/10/23 |

**MBL QA/QC's Representative:**

| | | | | | |
|---|---|---|---|---|---|
| **Name:** | Khaled Naim Sabha | **Signature:** | | **Date:** | 09/10/23 |

**Supervision Consultant's Review:**

**Checks:**
Site Test Results (if any): ☐ Passed ☐ Failed ☐ Not Applicable

**Status of Inspection:** ☑ Status A (Approved) ☐ Status B (Approved with Comments) ☐ Status C (Objection)

**Comments:** See attached comments.

Ramon
10/10/2023

| | | | | | |
|---|---|---|---|---|---|
| **Reviewer Name:** | Laxmi Krishna | **Signature:** | | **Date:** | 10/10/2023 |

**PMCM's Review:**

**Endorsement:** ☑ Yes ☐ No

**Comments:**

☐ To be Re-Submit

| | | | | | |
|---|---|---|---|---|---|
| **Reviewer Name:** | Fida | **Signature:** | | **Date:** | 12-10-2023 |

*Note: This site inspection does not relieve the contractor from any of his contractual obligations.*

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## Comments attach with (WIR)

WIR No. **L01-MBLC-CON-ELE-WIR-004002-R00** Dated: 10-10-2023

**Status - Code B**

**Inspection were done assuming that there will be no future changes it will be the contractor's sole responsibility to rectify the works further in case of any missing work or quality control requirements due to:**

- Installation are as per approved D04/IFC.
- Provided extra wires on boxes & panel boards shall be protected to all forms of damages.
- Contractor to ensure, provided extra wires is enough to final connection of wiring devices, Splice/joints on junction boxes will not be accepted.
- Insulation Resistance & Continuity test to be performed prior to termination of wires to panel board and wiring devices.
- Change of circuit-color/circuit no's to be updated in As-built drawings & load schedule
- The contractor is responsible for any future changes in terms of MEP coordination, D05 & final handover of the complete system.
- Quantity mentioned on WIR description is contractor responsibility.

**CSC Inspection:**

Name: Ramon B. Benedicto

Signature:

4002

| Documents Checklist Z15 SPS Second Fix | | | | |
|---|---|---|---|---|
| Description | Documents reference Number | Approved | Not Approved | UR |
| MES | L01-MBLC-CON-ELE-MES-0124-R02 | ☑ | ☐ | ☐ |
| ITP | L01-MBLC-CON-ELE-ITP-000724-R01 | ☑ | ☐ | ☐ |
| MAT | | ☐ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| | L01-MBLC-CON-ELE-MAT-0300-R00 | ☑ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| MIRs | L01-MBLC-CON-ELE-MIR-000564-R00 | ☑ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| Drawing Numbers | L01-MBLC-D04-SPS-DWG-10BD102-01-LAY402_R08 | ☑ | ☐ | ☐ |
| | | ☐ | ☐ | ☐ |
| | | | | |
| previous WIR Numbers | | | | |
| | L01-MBLC-CON-ELE-WIR- -R00 | ☐ | ☐ | ☑ |
| | | ☐ | ☐ | ☐ |

| | | | | |
|---|---|---|---|---|
| Location: | | Date: | | |
| Checklist Reference: | L01-MBLC-CON-ELE-CKT-000XXX-R0X | Reference ITP No. | | L01-FFIX-CON-ELE-ITP-0000XX-R0X |

Inspection / Witness by:

☐ FF Engineer ☐ MBLC ☐ SAPL/DARH ☐ CPPA ☐ Third Party

Checklist Tittle:

## Wire and Cable Pulling Checklist

| NO | Description | Not Applicable (NA) | Failed | Accepted |
|---|---|---|---|---|
| 1. | Material used are new and approved materials / wires & accessories | ☐ | ☐ | ☑ |
| 2. | Approved and stamped relevant IFC / Shop drawing & details are attached. | ☐ | ☐ | ☑ |
| 3. | Installation are as per relevant part of project specification. | ☐ | ☐ | ☑ |
| 4. | Work done as per approved method of statements. | ☐ | ☐ | ☑ |
| 5. | Store location and storage condition of cable/wires reels as per manufacturer recommendation/ instruction. | ☐ | ☐ | ☑ |
| 6. | Approved and certified tools, materials are used for pulling. | ☐ | ☐ | ☑ |
| 7. | Verify the exact route for pulled wires from Starts to End as per IFC/Shop drawings. | ☐ | ☐ | ☑ |
| 8. | Ensure LS0H wires (circuits & homeruns) are installed at plenum areas | ☐ | ☐ | ☑ |
| 9. | Correct phase marking and color coding arrangements are done as per Approved IFC and Shop drawings. | ☐ | ☐ | ☑ |
| 10. | Wires Type, Size and rating are used as per Approved IFC and Shop Drawings. | ☐ | ☐ | ☑ |
| 11. | Wires pulled are free from damages, twist and no visible scratches / cuts are found on the cable insulation. | ☐ | ☐ | ☑ |
| 12. | Verify the maximum filling ratio for conduit 40% or as per Specification. | ☐ | ☐ | ☑ |
| 13. | Check for extra length of the wires kept at either ends and ensured enough for terminations to intended devices/fixtures | ☐ | ☐ | ☑ |
| 14. | Ensure all required wires (phases, neutrals, groundings/bonding, controls etc.) are pulled as per IFC/Shop drawings. | ☐ | ☐ | ☑ |
| 15. | Verify all wiring routes, connection and terminations. ITBC | ☐ | ☐ | ☐ |
| 16. | Check proper terminals and lugs are used as per crimping Requirements. ITBC | ☐ | ☐ | ☐ |

| | | | | |
|---|---|---|---|---|
| Location: | | Date: | | |
| Checklist Reference: | L01-MBLC-CON-ELE-CKT-000XXX-R0X | Reference ITP No. | | L01-FFIX-CON-ELE-ITP-0000XX-R0X |

Inspection / Witness by:

☐ FF Engineer ☐ MBLC ☐ SAPL/DARH ☐ CPPA ☐ Third Party

Checklist Tittle:

**Wire and Cable Pulling Checklist**

| NO | Description | Not Applicable (NA) | Failed | Accepted |
|---|---|---|---|---|
| 17. | Prior to energization, test wires for continuity and IR Test. Results to be recorded on Test Form. ITBC | ☐ | ☐ | ☐ |
| 18. | Verify all grounding arrangement. ITBC | ☐ | ☐ | ☐ |
| 19. | Wire are pulled with appropriate lubricant. | ☐ | ☐ | ☑ |
| 20. | Required identifications for branch circuit wiring are done as per project specifications. | ☐ | ☐ | ☑ |
| 21 | Check for overall workmanship | ☐ | ☐ | ☑ |
| 22 | Others | ☐ | ☐ | ☑ |

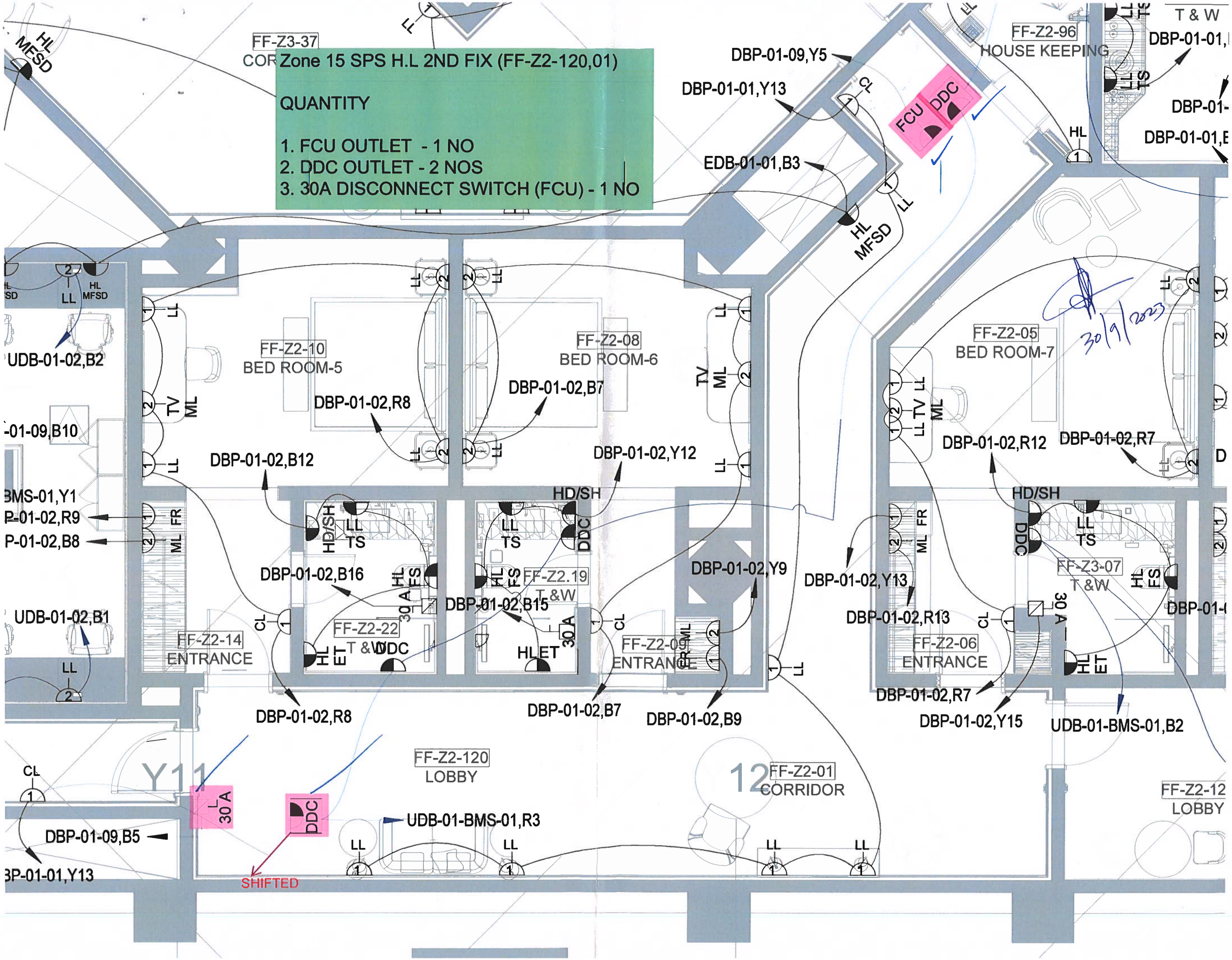
| Inspection / Test by: | | Witness and Approved: | |
|---|---|---|---|
| FIRST FIX OR VENDOR : | QC Engineer: | MBLC | CSC |
| First Fix | SM Junaid 3/10/23 | 3-10-23 Abdulkerim | 10/10/2023 |
| NAME, SIGN & DATE | NAME, SIGN & DATE | NAME, SIGN & DATE | NAME, SIGN & DATE |

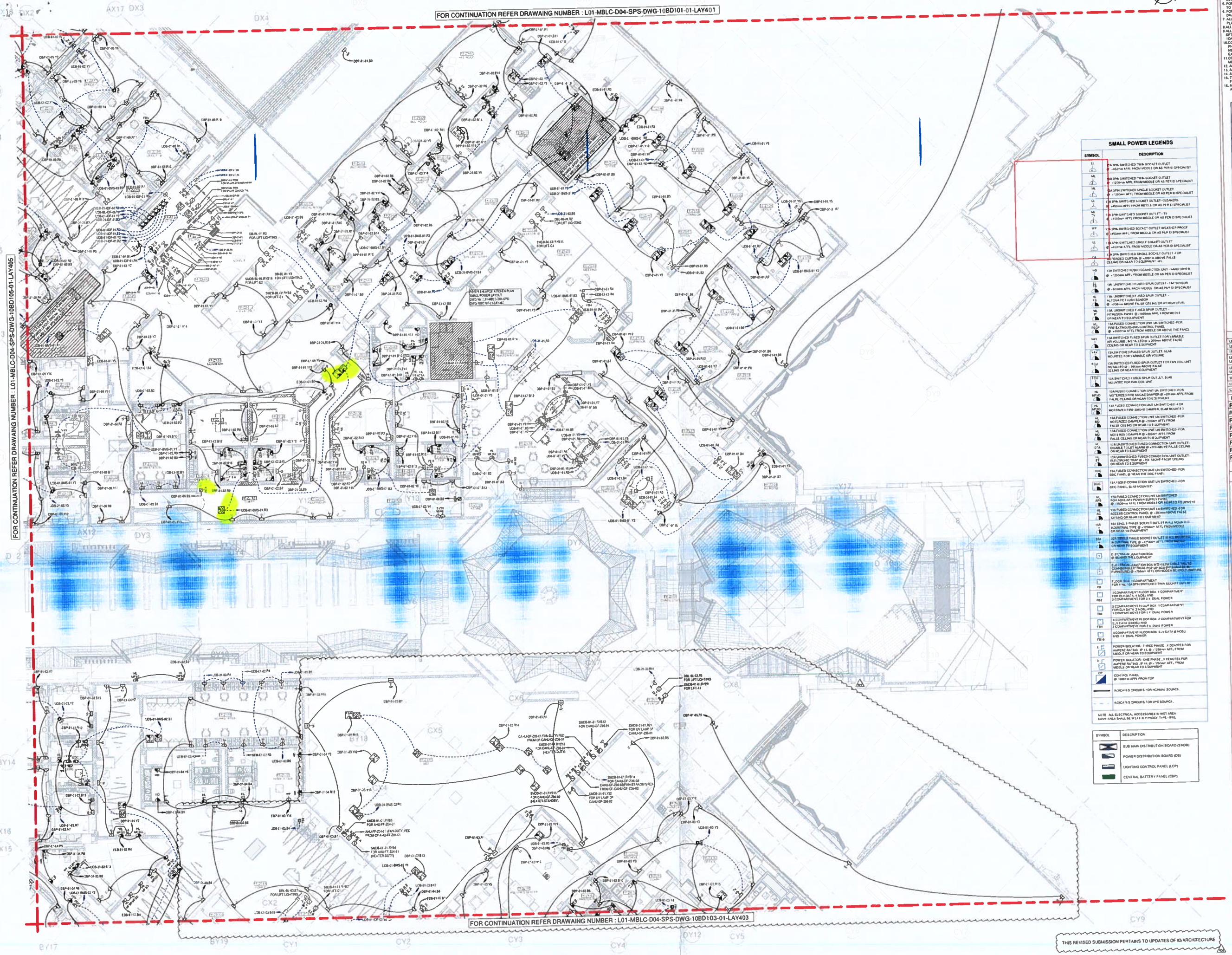
**Note: This checklist may be tailored / edited in-line with the actual works related to each WIR**

Zone 15 SPS H.L 2ND FIX (FF-Z2-120,01)

QUANTITY

1. FCU OUTLET - 1 NO
2. DDC OUTLET - 2 NOS
3. 30A DISCONNECT SWITCH (FCU) - 1 NO






1 **03-FF - First Floor Plan Power Layout - Part-2**
SCALE - 1 : 100

THIS REVISED SUBMISSION PERTAINS TO UPDATES OF IDI ARCHITECTURE

## GENERAL NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS AND LEVELS ARE IN METERS UNLESS NOTED OTHERWISE.
2. DO NOT SCALE FROM THIS DRAWING. ONLY WRITTEN DIMENSIONS ARE TO BE FOLLOWED
3. THE DRAWINGS SHALL BE READ IN COLLABORATION WITH ALL RELEVANT DOCUMENTS AND PROJECT SPECIFICATIONS.
4. INSTALLATIONS IN ALL OUTDOOR AND WET AREAS SHALL BE WEATHERPROOF TYPE & IP 65/66 RATING.
5. FOR EXACT LOCATION AND POWER CONSUMPTION OF THE IT AND TELECOM LOADS, REFER TO THE ELV AND TELECOM DRAWINGS AND SCHEDULES.
6. FOR EXACT LOCATION AND POWER CONSUMPTION OF THE HVAC LOADS REFER TO HVAC DRAWINGS AND SCHEDULES.
7. ALL POWER OUTLETS LOCATIONS SHALL BE COORDINATED WITH THE FINAL FURNITURE PLANS.
8. ALL SMALL POWER AND LIGHTING DISTRIBUTION BOARD TO BE FORM 2B TYPE 2.
9. ALL RATINGS ARE NOMINAL. CONTRACTOR IS RESPONSIBLE FOR FINAL RATINGS AND SETTINGS AFTER A THOROUGH PREPARATION OF SHOP DRAWINGS CONSIDERING ACTUAL LOADS AND ADHERE TO LOCAL REGULATIONS.
10. COLOUR CODING FOR CIRCUITS SHALL BE AS FOLLOWS:
    PHASES : RED, YELLOW, BLUE
    NEUTRAL : BLACK
    EARTHING : YELLOW GREEN
11. CONDUIT SIZE FOR EMBEDDED AND SURFACE INSTALLATION SHALL FOLLOW LAYOUTS, MINIMUM 20MM SIZE TO BE FOLLOWED.
12. ALL 3 PHASE ISOLATORS ARE WITH 3POLE UNLESS OTHERWISE INDICATED.
13. ALL 1 PHASE ISOLATORS ARE WITH 1POLE UNLESS OTHERWISE INDICATED
14. ALL WIRINGS FOR SMALL POWER SHALL NOT BE SMALLER THAN 4.0 mm².
15. THE LAYOUT IS DIAGRAMMATIC AND ACTUAL FEEDER ROUTING SHALL USE SHORTEST PATH TO CIRCUIT SOURCE.
16. ALL CIRCUITS LOCATED IN WET AREAS SHALL BE PROVIDED WITH RCBO OR ELCB WITH APPROPRIATE RATING.

## SMALL POWER LEGENDS

| SYMBOL | DESCRIPTION |
|---|---|
| LL | 13A SPN SWITCHED TWIN SOCKET OUTLET @ +300mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| ML | 13A SPN SWITCHED TWIN SOCKET OUTLET @ +1200mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| HL | 13A SPN SWITCHED SINGLE SOCKET OUTLET @ +2000mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| CL | 13A SPN SWITCHED SOCKET OUTLET- CLEANERS @ +400mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| ML TV | 13A SPN SWITCHED SOCKET OUTLET - TV @ +1200mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| WP | 13A SPN SWITCHED SOCKET OUTLET-WEATHER PROOF @ +450mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| LL | 13A SPN SWITCHED SINGLE SOCKET OUTLET @ +300mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| CA | 13A SPN SWITCHED SINGLE SOCKET OUTLET - FOR MOTORIZED CURTAIN @ +2400mm ABOVE FALSE CEILING OR NEAR TO EQUIPMENT. MIL |
| HD | 13A SWITCHED FUSED CONNECTION UNIT - HAND DRYER @ +1500mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| | 13A UNSWITCHED FUSED SPUR OUTLET - TAP SENSOR @ +600mm AFFL FROM MIDDLE OR AS PER ID SPECIALIST |
| HL FS | 13A UNSWITCHED FUSED SPUR OUTLET - AUTOMATIC FLUSH SENSOR @ +2300mm ABOVE FALSE CEILING OR AT HIGH LEVEL |
| ML P | 13A UNSWITCHED FUSED SPUR OUTLET - INTERCOM PANEL @ +1600mm AFFL FROM MIDDLE OR NEAR TO EQUIPMENT |
| KL FCUP | 13A FUSED CONNECTION UNIT UN SWITCHED -FOR FIRE EXTINGUISHING CONTROL PANEL @ +1500mm AFFL FROM MIDDLE OR ABOVE THE PANEL |
| HAV | 13A SWITCHED FUSED SPUR OUTLET FOR VARIABLE AIR VOLUME INSTALLED IN +2000mm ABOVE FALSE CEILING OR NEAR TO EQUIPMENT |
| HAV FCU | 13A SWITCHED FUSED SPUR OUTLET, SLAB MOUNTED FOR VARIABLE AIR VOLUME |
| | 13A SWITCHED FUSED SPUR OUTLET FOR FAN COIL UNIT INSTALLED @ +2400mm ABOVE FALSE CEILING OR NEAR TO EQUIPMENT |
| FCU | 13A SWITCHED FUSED SPUR OUTLET, SLAB MOUNTED FOR FAN COIL UNIT |
| HL MFD | 13A FUSED CONNECTION UNIT UN SWITCHED -FOR MOTORIZED FIRE SMOKE DAMPER @ +2000mm AFFL FROM FALSE CEILING OR NEAR TO EQUIPMENT |
| HL MFD SD | 13A FUSED CONNECTION UNIT UN SWITCHED -FOR MOTORIZED FIRE SMOKE DAMPER, SLAB MOUNTED |
| HL MD | 13A FUSED CONNECTION UNIT UN SWITCHED -FOR MOTORIZED DAMPER @ +2000mm AFFL FROM FALSE CEILING OR NEAR TO EQUIPMENT |
| HL MD SD | 13A FUSED CONNECTION UNIT UN SWITCHED -FOR MOTORIZED DAMPER @ +2000mm AFFL FROM FALSE CEILING OR NEAR TO EQUIPMENT |
| HL DTA | 13A UNSWITCHED FUSED CONNECTION UNIT OUTLET- DISABLE TOILET ALARM @ +400 ABOVE FALSE CEILING OR NEAR TO EQUIPMENT |
| HL FT | 13A UNSWITCHED FUSED CONNECTION UNIT OUTLET FLE STROBE TRAP @ +400 ABOVE FALSE CEILING OR NEAR TO EQUIPMENT |
| DDC | 13A FUSED CONNECTION UNIT UN SWITCHED -FOR DDC PANEL @ NEAR THE DDC PANEL |
| DDC | 13A FUSED CONNECTION UNIT UN SWITCHED -FOR DDC PANEL, SLAB MOUNTED |
| ML AP | 13A FUSED CONNECTION UNIT UN SWITCHED FOR ACCESS APU POWER SUPPLY PANEL @ +1500mm AFFL FROM MIDDLE OR NEAR TO EQUIPMENT |
| HL AC | 13A FUSED CONNECTION UNIT UNSWITCHED FOR ACCESS CONTROL PANEL @ +2000mm ABOVE FALSE CEILING OR NEAR TO EQUIPMENT |
| 16A | 16A SING. PHASE SOCKET OUTLET IN WALL MOUNTED INDUSTRIAL TYPE @ +1200mm AFFL FROM MIDDLE OR NEAR TO EQUIPMENT |
| 32A | 32A SINGLE PHASE SOCKET OUTLET IN WALL MOUNTED INDUSTRIAL TYPE @ +1200mm AFFL FROM MIDDLE OR NEAR TO EQUIPMENT |
| | ELECTRICAL JUNCTION BOX @ BELOW FINAL EQUIPMENT |
| | ELECTRICAL JUNCTION BOX WITH FLEXIBLE CABLE TAIL TO CONNECT ELECTRICAL PLUG UP FOR HIDDEN BUILT-IN EQUIPMENT FURNITURE @ +750mm AFFL FOR HIDDEN BUILT-IN FURNITURE |
| FB | FLOOR BOX 1 COMPARTMENT FOR 1 No. 13A SPN SWITCHED TWIN SOCKET OUTLET |
| FB2 | 2 COMPARTMENT FLOOR BOX 1 COMPARTMENT FOR ELV DATA & HDSL AND 1 COMPARTMENT FOR 1 X DUAL POWER |
| FB4 | 3 COMPARTMENT FLOOR BOX 1 COMPARTMENT FOR ELV DATA & 2 HDSL AND 1 COMPARTMENT FOR 1 X DUAL POWER |
| FB6 | 4 COMPARTMENT FLOOR BOX 2 COMPARTMENT FOR ELV DATA (BNDSL) AND 2 COMPARTMENT FOR 2 X DUAL POWER |
| FB10 | 4 COMPARTMENT FLOOR BOX ELV DATA (8 HDSL) AND 1 X DUAL POWER |
| | POWER ISOLATOR - THREE PHASE , X DENOTES FOR AMPERE RATING. IP 66 @ +250mm AFFL FROM MIDDLE OR NEAR TO EQUIPMENT |
| | POWER ISOLATOR - ONE PHASE , X DENOTES FOR AMPERE RATING. IP 66 @ +250mm AFFL FROM MIDDLE OR NEAR TO EQUIPMENT |
| CP | CONTROL PANEL @ 1600mm AFFL FROM TOP |
| | INDICATES CIRCUITS FOR NORMAL SOURCE. |
| | INDICATES CIRCUITS FOR UPS SOURCE. |

NOTE: ALL ELECTRICAL ACCESSORIES IN WET AREA DAMP AREA SHALL BE WEATHER PROOF TYPE - IP65.

| SYMBOL | DESCRIPTION |
|---|---|
| | SUB MAIN DISTRIBUTION BOARD (SMDB) |
| | POWER DISTRIBUTION BOARD (DB) |
| | LIGHTING CONTROL PANEL (LCP) |
| | CENTRAL BATTERY PANEL (CBP) |

**SAPL**
SAUDI ARABIAN PARSONS LTD
شركة بارسونز العربية السعودية المحدودة

- [ ] 1 - No Objection
- [x] 2 - No Objection with Comments
- [ ] 3 - Objection

It is the Contractor's sole responsibility to ensure complete coordination with no clashes among the relevant disciplines. No Objection does not relieve the D&B Contractor of his contractual obligations to comply with the specified standards, local regulations, and Employer's requirements.

The Engineer considers his review a notice to correct by the Contractor as per clause 5.2 of the Contract.

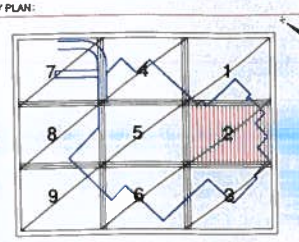
Name: Date:

| DRAWING NUMBER | REFERENCE NAME |
|---|---|
| L01-MBLC-D04-ELE-DWG-10BC100-GN-DETB01 | ELECTRICAL LEGEND & ABBREVIATIONS |
| L01-MBLC-D04-ELE-DWG-10BC100-GN-DETB02 | ELECTRICAL GENERAL NOTES |
| L01-MBLC-D04-SPS-DWG-10BD100-GN-DETB01 | ELECTRICAL SMALL POWER TYPICAL DETAILS |
| L01-MBLC-D04-LVS-DWG-10BE100-ZZ-SC-D76-TC | LV SWITCHGEAR POWER SCHEMATIC DIAGRAMS |
| L01-MBLC-D04-LVS-DWG-10BE100-ZZ-SC-D110-TC<br>L01-MBLC-D04-LVS-DWG-10BE100-ZZ-SC-D116 | LV POWER SCHEMATIC DIAGRAMS |
| L01-JAPN-GEN-ELE-RPT-1380 | FIRST LEVEL DB SCHEDULE REPORT |
| L01-MBLC-D04-MHV-DWG-10BD101-01-LAY402 | FIRST FLOOR PART 2 - HVAC DUCTING LAYOUT |
| L01-MBLC-D04-DRN-DWG-10BD101-01-LAY402 | FIRST FLOOR PART 2 - DRAINAGE LAYOUT |

REVISIONS:

| NO. | DATE | DESCRIPTION | APPROVED |
|---|---|---|---|
| R08 | 27/03/2023 | ISSUED FOR CONSTRUCTION | AJ |
| R07 | 20/03/2023 | ISSUED FOR CONSTRUCTION | AJ |
| R06 | 02/01/2023 | ISSUED FOR CONSTRUCTION | AJ |
| R05 | 23/12/2022 | ISSUED FOR CONSTRUCTION | AJ |
| R04 | 03/11/2022 | ISSUED FOR CONSTRUCTION | AJ |
| R03 | 17/08/2022 | ISSUED FOR CONSTRUCTION | AJ |
| R02 | 01/08/2021 | ISSUED FOR CONSTRUCTION | AJ |
| R01 | 21/11/2020 | ISSUED FOR CONSTRUCTION | AJ |

KEY PLAN:




PROJECT MANAGEMENT CONSULTANT:

PARSONS

MAIN CONTRACTOR:

MBL

DESIGNER:

Jain

BARC SOLUTIONS

TXG

PROJECT TITLE:

L01
Riyadh - Saudi Arabia

DRAWING TITLE:

FIRST FLOOR PART 2
SMALL POWER LAYOUT

| JOB NO. | ACTUAL SIZE | DRAWN | CHECKED |
|---|---|---|---|
| L01 | A0 | MJ | SVR |
| | SCALE | DESIGNED | APPROVED |
| DRAWING NO. | 1 : 100 | SVR | AJ |

L01-MBLC-D04-SPS-DWG-10BD102-01-LAY402

| STATUS | DATE | REVISION |
|---|---|---|
| ISSUED FOR CONSTRUCTION | 27/03/2023 | R08 |